

Investment memo: 50 MW ground-mounted solar PV, Germany

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Project finance screening | Sources: PVGIS, SMARD (Bundesnetzagentur), Fraunhofer ISE, Bundesnetzagentur tender results

Recommendation: do not invest on current terms. Proceed only if revenue of about 56 EUR/MWh or more can be secured at a southern site with CAPEX near 700 EUR/kWp.

At the 2025 solar capture price (46.1 EUR/MWh) and the July 2026 average EEG tender award (47.9 EUR/MWh), no scenario in the model reaches the 6.5% equity return used by Fraunhofer ISE. The best case (south, CAPEX 700) earns an equity IRR of 3.1% under an EEG award. The conditions that would change the decision:

- **Price:** a flat-equivalent revenue of at least 56 EUR/MWh (south, CAPEX 700), 63 (south, CAPEX 800) or 65 (Central/East, CAPEX 700), i.e. 8.5 to 17.4 EUR/MWh above the tender level.
- **CAPEX:** at the low end of the published range. Each additional 100 EUR/kWp raises the required price by about 7 EUR/MWh.
- **Site:** a southern yield of about 1,215 kWh/kWp or better. The Central/East site needs about 9 EUR/MWh more.
- **Verification before any commitment:** site-specific PVGIS yield, contractor CAPEX quotes, a lender term sheet, and the EEG treatment of negative-price hours.

1. Project and approach

A 50 MW ground-mounted PV plant with a 30-year life. The Python model builds annual cash flows (generation with degradation, revenue, OPEX, tax), sizes debt to a target DSCR, and returns project IRR, equity IRR and LCOE. Two revenue routes are compared: **merchant** sales at the solar capture price for 30 years, and an **EEG award** of 47.9 EUR/MWh for 20 years followed by merchant sales. **Validation:** run with Fraunhofer ISE's own inputs, the model gives a generation cost of 4.13 ct/kWh (south, CAPEX 700) and 6.86 ct/kWh (north, CAPEX 900), matching the study's published range of 4.1 to 6.9 ct/kWh.

Input	Value	Source
Specific yield	Central/East 1,049; South about 1,215 kWh/kWp	PVGIS-SARAH3 (52.5 N, 13.4 E, 35 deg, 14% losses); South scaled by 1,280/1,105 from Fraunhofer ISE Table 3 (approximation)
CAPEX / OPEX	700, 800, 900 EUR/kWp / 13.3 EUR/kW/yr	Fraunhofer ISE (2024), Tables 1 and 2
Lifetime, degradation	30 years, 0.25% per year	Fraunhofer ISE (2024), Table 2
Debt, equity benchmark	5.0% interest, max 80% gearing; 6.5% equity return	Fraunhofer ISE (2024), Table 2
Solar capture price	46.1 EUR/MWh (2025); 46.2 (2024)	SMARD day-ahead DE-LU weighted by SMARD solar generation
EEG tender award	47.9 EUR/MWh	Bundesnetzagentur, ground-mounted solar, 1 July 2026
Tax, DSCR, tenor	30%, 1.25x, 18 years	Placeholders (tested in section 2)

2. Results

Site	CAPEX (EUR/kWp)	Equity IRR, merchant	Equity IRR, EEG award	Price for 6.5% equity (EUR/MWh)
Central/East	700	0.4%	0.8%	65.3
Central/East	800	-0.7%	-0.4%	72.8
Central/East	900	-1.7%	-1.4%	80.3
South	700	2.6%	3.1%	56.4
South	800	1.1%	1.5%	62.8
South	900	0.0%	0.3%	69.3

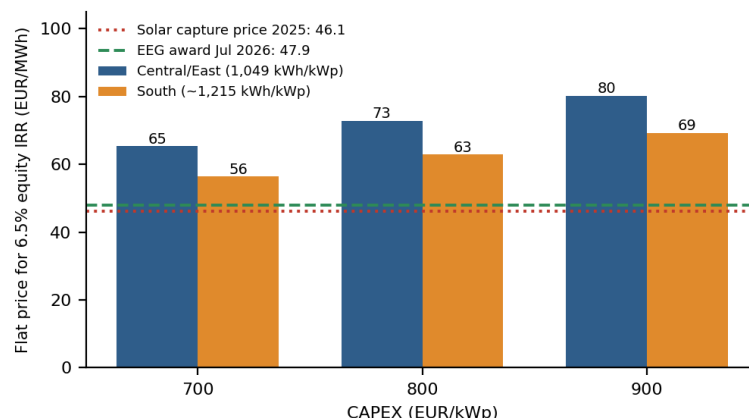


Figure: flat power price needed for a 6.5% equity IRR, by site and CAPEX, against the two reference prices.

- Site yield is worth about 9 EUR/MWh and each +100 EUR/kWp of CAPEX about 7 EUR/MWh in required price. Even the best case is about 8.5 EUR/MWh above the tender level.
- **Financing is not the lever.** Lowering the DSCR from 1.25x to 1.15x and extending the tenor from 18 to 20 years moves the required price by at most 1.5 EUR/MWh. The project IRR (about 3.5% in the best case) is below the 5% cost of debt, so extra

leverage does not raise equity returns.

- **Inflation:** with 1.8% annual OPEX inflation and a fixed nominal award, the best case falls from 3.1% to 2.1% equity IRR.

3. Key risks

- **Price cannibalisation.** From SMARD data: baseload rose from 78.5 (2024) to 89.3 EUR/MWh (2025), yet the solar capture price stayed at about 46 EUR/MWh. The capture rate fell from 59% to 52%. Hours with negative prices rose from 457 to 573, and the share of solar output sold in those hours rose from 18% to 24%. In May and June 2025 solar earned only about 20 to 22 EUR/MWh.
- **CAPEX and site.** The published CAPEX range (700 to 900 EUR/kWp) alone moves the required price by about 14 EUR/MWh. The yield difference between the two sites is worth about 9 EUR/MWh.
- **Policy and market design.** The treatment of negative-price hours and the length of support under the EEG are not modelled and must be checked for the specific tender round.
- **Negative leverage.** Because project returns are below the cost of debt, higher gearing increases equity risk without raising the expected return.

4. Limitations and next steps

Limitations. Flat nominal prices (only OPEX inflation is tested); simplified tax (30%, 20-year straight-line depreciation, no loss carry-forward); no reserve accounts or construction period; the southern yield is a scaled approximation; the capture price is a national 2025 day-ahead value from two years of data; the 6.5% equity return is a study assumption, not a bidder's hurdle rate; the EEG award is a national average, not a project-specific bid. Developers do win tenders at these prices, so real projects may rely on lower CAPEX, higher-yield designs (such as trackers or bifacial modules), lower return requirements or financing terms that this model does not capture.

Next steps. (1) Run PVGIS for specific southern sites. (2) Replace the CAPEX range with contractor quotes. (3) Add lender terms and a forward price curve. (4) Test PV plus battery storage as a way to lift the capture price. (5) Re-run the Monte Carlo risk analysis with the sourced inputs.